

ภาคผนวก ข

Function in Libpcap which used in this project.

pcap_open_offline () is called to open a ``savefile" for reading. fname specifies the name of the file to open. The file has the same format as those used by `tcpdump(1)` and `tcpdump(1)`. The name "-" is a synonym for stdin. errbuf is used to return error text and is only set when `pcap_open_offline()` fails and returns NULL.

pcap_loop () is similar to `pcap_dispatch()` except it keeps reading packets until cnt packets are processed or an error occurs. It does not return when live read timeouts occur. Rather, specifying a non-zero read timeout to `pcap_open_live()` and then calling `pcap_dispatch()` allows the reception and processing of any packets that arrive when the timeout occurs. A negative cnt causes `pcap_loop()` to loop forever (or at least until an error occurs). -1 is returned on an error; 0 is returned if cnt is exhausted; -2 is returned if the loop terminated due to a call to `pcap_breakloop()` before any packets were processed. If your application uses `pcap_breakloop()`, make sure that you explicitly check for -1 and -2, rather than just checking for a return value < 0.

pcap_dump() outputs a packet to the ``savefile" opened with `pcap_dump_open()`. Note that its calling arguments are suitable for use with `pcap_dispatch()` or `pcap_loop()`. If called directly, the user parameter is of type `pcap_dumper_t` as returned by `pcap_dump_open()`.

pcap_dump_file() returns the standard I/O stream of the ``savefile" opened by `pcap_dump_open()`.

pcap_open_live() is used to obtain a packet capture descriptor to look at packets on the network. device is a string that specifies the network device to open; on Linux systems with 2.2 or later kernels, a device argument of "any" or NULL can be used to capture packets from all interfaces. snaplen specifies the maximum number of bytes to capture. If this value is less than the size of a packet that is captured, only the first snaplen bytes of that packet will be captured and provided as packet data. A value of 65535 should be sufficient, on most if not all networks, to capture all the data available from the packet. promisc specifies if the interface is to be put into promiscuous mode. (Note that even if this parameter is false, the interface could well be in promiscuous mode for some other reason.) For now, this doesn't work on the "any" device; if an argument of "any" or NULL is supplied, the promisc flag is ignored. to_ms specifies the read timeout in milliseconds. The read timeout is used to arrange that the read not necessarily return immediately when a packet is seen, but that it wait for some amount of time to allow more packets to arrive and to read multiple packets from the OS kernel in one operation. Not all platforms support a read timeout; on platforms that don't, the read timeout is ignored. A zero value for to_ms, on platforms that support a read timeout, will cause a read to wait forever to allow enough packets to arrive, with no timeout. errbuf is used to return error or warning text. It will be set to error text when `pcap_open_live()` fails and returns NULL. errbuf may also be set to warning text when `pcap_open_live()` succeeds; to detect this case the caller should store a zero-length string in errbuf before calling `pcap_open_live()` and display the warning to the user if errbuf is no longer a zero-length string.

pcap_freealldevs() is used to free a list allocated by `pcap_findalldevs()`.

pcap_findalldevs() constructs a list of network devices that can be opened with `pcap_open_live()`. (Note that there may be network devices that cannot be opened with

`pcap_open_live()` by the process calling `pcap_findalldevs()`, because, for example, that process might not have sufficient privileges to open them for capturing; if so, those devices will not appear on the list.) `alldevsp` is set to point to the first element of the list; each element of the list is of type `pcap_if_t`, and has the following members:

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